

# Waqeel AI

## Pakistan Ka Digital Waqeel — Har Gareeb Ka Haq Hai

### AI-Powered Free Legal Guidance System for Pakistani Citizens

Voice-First | Urdu Language | RAG-Based | Free for All Citizens

220M+

Potential Users

7 Law Domains

Phase 1 Coverage

100%

Free for Citizens

Voice First

Urdu Interface

|                |                                                   |
|----------------|---------------------------------------------------|
| Project Title  | Waqeel AI — Pakistan Ka Digital Legal Guide       |
| Domain         | LegalTech / AI / NLP / Voice Agents               |
| Target Users   | Pakistani citizens, especially low-income & rural |
| Core Tech      | FastAPI · LangChain · Whisper · RAG · Mistral AI  |
| Languages      | Urdu · Roman Urdu · Punjabi · Sindhi (Phase 2)    |
| Business Model | Free for citizens · Grant/NGO · Lawyer referrals  |
| Document       | v1.0 — Client Presentation Documentation          |

## 1. Problem Statement

Pakistan mein 220 million+ ki abadi mein se estimated 60% legally illiterate hai — yani unhe apne basic qanooni haqooq ka ilm nahi. Yeh sirf ek education gap nahi, yeh ek systemic failure hai jo rozmarra ke zindagi mein direct financial aur social nuksan deta hai.

|                                                |                                                                                                    |
|------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Scenario: Landlord ne bina notice ke nikaala   | Result: Tenant ko pata nahi tha ke 30-din ka written notice legally required tha. Ghar chhor diya. |
| Scenario: Traffic police ne galat chalaan diya | Result: Driver ko maloom nahi tha kaunse qaida ka hawaala diya gaya, kisi ne challenge nahi kiya.  |

|                                                    |                                                                                          |
|----------------------------------------------------|------------------------------------------------------------------------------------------|
| Scenario: Employer ne without cause terminate kiya | Result: Worker ko nahi pata tha NIRC mein 30-din ka haq milta hai, koi action nahi liya. |
|----------------------------------------------------|------------------------------------------------------------------------------------------|

Root Causes

- **Legal illiteracy:** Laws complex language mein hain, Urdu translation nahi ya inaccessible hai.
- **Lawyer affordability:** Basic consultation PKR 5,000–50,000+ — majority afford nahi kar sakti.
- **Information scattered:** Koi ek centralized, reliable, free source nahi Pakistan mein.
- **Language barrier:** Most online resources English mein hain.
- **No voice option:** Low-literacy users text-based apps use nahi kar sakte.

Existing Gaps

| Solution         | What it does                          | What's Missing                                |
|------------------|---------------------------------------|-----------------------------------------------|
| Google/websites  | General legal info in English         | Not Pakistan-specific, not voice, not Urdu    |
| Lawyers          | Personalized legal advice             | Expensive (PKR 5,000-50,000+), not accessible |
| Legal aid NGOs   | Free advice at physical offices       | Limited hours, limited cities, not scalable   |
| Zameen/Bayut     | Property listings                     | Not a legal guide                             |
| Waqeel AI (Ours) | Free, voice, Urdu, Pakistan law, 24/7 | ✓ This gap filled by our solution             |

## 2. Solution Overview — Waqeel AI

Waqeel AI ek free, voice-first, AI-powered legal guidance system hai jo Pakistani citizens ko unke qanooni haqooq ke baare mein Urdu mein guide karta hai — 24/7, anonymously, bina kisi fees ke.

|   |                                                                                                         |
|---|---------------------------------------------------------------------------------------------------------|
| 1 | <b>User bolta hai</b><br>Urdu ya Roman Urdu mein apni situation describe karta hai — voice ya text mein |
| 2 | <b>AI samajhta hai</b><br>Speech-to-text (Whisper) se convert, NLP se intent identify                   |
| 3 | <b>Law dhundta hai</b><br>RAG engine Pakistan ke relevant laws se exact sections nikalta hai            |
| 4 | <b>Guide karta hai</b><br>Simple Urdu mein explanation + voice output + next steps                      |
| 5 | <b>Document banata hai</b><br>Zarurat ho toh FIR draft, legal notice ya complaint ready                 |
| 6 | <b>Lawyer suggest karta hai</b><br>Complex cases mein nearest free legal aid refer karta hai            |

### Real Example Conversation

|            |                                                                                                                                                                                                                                                                    |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| USER:      | Mere makan maalik ne mujhe kal tak ghar khaali karne ko kaha hai, koi notice nahi diya.                                                                                                                                                                            |
| WAQEEL AI: | Pakistan Rent Restriction Ordinance 1949 ke Section 6 ke mutabiq, koi bhi landlord aapko bina 30 din ka written notice diye nahi nikaal sakta. Agar wo aisa kare toh yeh qanoonan illegal hai. Kya main aapke liye City Court mein complaint ka draft tayar karun? |
| USER:      | Haan, draft karo.                                                                                                                                                                                                                                                  |
| WAQEEL AI: | Document tayar ho gaya hai. Apna naam aur address fill karein aur nearest City Court mein submit karein. Kya aapko nearest court ka address chahiye?                                                                                                               |

### 3. Core Features

VOICE

#### 3.1 Voice-First Urdu Interface

User microphone se bolta hai — ya type karta hai — Urdu, Roman Urdu, ya regional language mein. System speech recognize karta hai, samajhta hai, aur voice mein jawaab deta hai. Low-literacy users ke liye specifically designed.

- Urdu speech recognition (OpenAI Whisper)
- Roman Urdu understanding (normalized input layer)
- Text-to-speech Urdu output (gTTS / ElevenLabs)
- Text fallback always available
- Works on basic Android phones

RAG

#### 3.2 Pakistan Law RAG Engine

Retrieval-Augmented Generation (RAG) engine Pakistan ki actual law books se specific sections dhundh kar nikalta hai — sirf general AI knowledge nahi, verified legal text pe grounded answers deta hai.

- Pakistan Penal Code (PPC) — crimes & punishments
- Code of Criminal Procedure (CrPC) — arrest, bail, FIR
- Rent Restriction Ordinance — tenant/landlord rights
- Muslim Family Laws — divorce, inheritance, nikah
- Consumer Protection Act — fraud, defective products
- Labor Laws — termination, wages, NIRC
- Motor Vehicle Ordinance — traffic fines, licenses

TRIAGE

#### 3.3 Smart Case Triage System

Har case ke baad system honestly assess karta hai aur citizen ko batata hai kitna serious hai unka masla — aur kya khud handle ho sakta hai ya lawyer zaruri hai.

- GREEN: Aap khud handle kar sakte hain (simple chalaan, consumer complaint)
- YELLOW: Ek baar lawyer se consult karein (property, labor dispute)
- RED: Foran lawyer zaruri (criminal case, serious charges, arrest)
- Nearest free legal aid automatically suggest hoti hai

DOCS

#### 3.4 Legal Document Draft Generator

System conversation ke baad automatically ready-to-submit legal documents draft karta hai — user sirf apna naam aur details fill kare.

- FIR draft (thana mein submit karne ke liye)
- Legal notice (landlord, employer, seller ko)
- Consumer complaint (Consumer Court ke liye)
- Khula application (basic)
- RTI (Right to Information) application
- Bail application (basic — critical cases mein lawyer refer)

ANON

### 3.5 Fully Anonymous Mode

Sensitive legal situations mein log identity expose karne se darte hain. Waqeel AI mein koi account, koi login, koi data storage nahi — session khatam, sab data delete.

- Zero account required
- No conversation logging (privacy-first)
- No CNIC or personal data collection
- Session-only memory (clears on exit)
- Open source code (community verified)

CARD

### 3.6 Offline Rights Emergency Card

User ek digital/printable card generate kar sakta hai jo offline bhi kaam kare — jeb mein rakho, zarurat pe padhlo. Internet nahi ho toh bhi kaam aaye.

- Arrest rights (CrPC 340 — lawyer ka haq)
- Bail rights summary
- Emergency helpline numbers
- Nearest free legal aid contacts
- Print-friendly format

## 4. Advanced Features

Yeh features Waqeel AI ko kisi bhi existing Pakistani platform se genuinely differentiate karte hain:

|                                   |                                                                                                                                                              |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multi-Language NLP                | Phase 1: Urdu + Roman Urdu. Phase 2: Punjabi, Sindhi, Pashto basic support. Koi bhi existing Pakistani legal tool yeh nahi karta.                            |
| Fake Document Detector            | FIR ya legal notice ki photo upload karo — AI check kare kaunse sections apply hote hain, kya koi suspicious clause hai, aur kya document genuine lagta hai. |
| Law Change Auto-Update            | Pakistan mein laws amend hoti rehti hain. System version-controlled law database maintain karta hai aur quarterly updates apply hoti hain.                   |
| Punishment & Fine Estimator       | User bataye kaunsa section laga hai — system estimate kare minimum aur maximum punishment, bail eligibility, aur typical timeline.                           |
| Lawyer Referral Network (Revenue) | Complex cases mein verified lawyers suggest ho — lawyer ko referral milta hai, platform ko commission. Users ke liye free rehta hai.                         |
| WhatsApp Integration              | SMS ya WhatsApp pe bhi system access ho jaye — smartphones without internet bhi use kar sakein basic queries ke liye.                                        |

## 5. Technical Architecture

[ User – Voice/Text Input (Mobile/Web) ] | | HTTPS / WhatsApp API | [ FastAPI Backend – Core Orchestration Layer ] | | | [ Voice [ RAG [ Document [ Triage Agent ] Engine ] Generator ] Classifier ] | | | [ Whisper ] [ FAISS [ Template [ ML Model ] + Mistral ] Engine ] | [ Pakistan Law PDF Database ] [ PPC | CrPC | Tenancy | Family | Labor ] | [ PostgreSQL ] [ Redis Cache ] [ Twilio/SMS ]

### Tech Stack

| Layer              | Tool                       | Why Chosen                                      |
|--------------------|----------------------------|-------------------------------------------------|
| Voice Input        | OpenAI Whisper             | Best Urdu speech recognition, runs locally free |
| Voice Output       | gTTS / ElevenLabs          | Natural Urdu TTS, gTTS free tier available      |
| Backend API        | FastAPI (Python)           | High performance, async, already skilled        |
| AI/LLM             | Mistral AI (mistral-small) | Already have API key, cost-effective            |
| RAG Framework      | LangChain                  | Document ingestion, retrieval pipeline          |
| Vector Database    | FAISS (local)              | Free, fast, runs without cloud dependency       |
| Law PDF Parsing    | pdfplumber + LangChain     | Extract structured text from law PDFs           |
| Document OCR       | Tesseract                  | Free, analyzes uploaded documents               |
| NLP/Classification | Scikit-learn               | Triage classifier, intent detection             |
| Database           | PostgreSQL                 | Reliable, free, structured storage              |
| Cache/Queue        | Redis + Celery             | Law update jobs, session management             |
| Frontend           | React.js / HTML+JS         | Mobile-first responsive interface               |
| Notifications      | Twilio                     | SMS fallback for no-internet users              |
| Hosting            | HuggingFace Spaces         | Free deployment, Docker support                 |

### Data Sources (All Free & Legal)

| Source              | URL                                                          | Laws Covered               |
|---------------------|--------------------------------------------------------------|----------------------------|
| Pakistan Law Site   | <a href="http://pakistanlawsite.com">pakistanlawsite.com</a> | PPC, CrPC, major acts      |
| Federal Legislative | <a href="http://na.gov.pk">na.gov.pk</a>                     | Official acts & amendments |

|               |                                                                                         |                            |
|---------------|-----------------------------------------------------------------------------------------|----------------------------|
| Punjab Laws   | <a href="http://punjablaws.gov.pk">punjablaws.gov.pk</a>                                | Punjab-specific ordinances |
| Sindh Laws    | <a href="http://sindhlaws.gov.pk">sindhlaws.gov.pk</a>                                  | Sindh-specific laws        |
| Supreme Court | <a href="http://supremecourt.gov.pk">supremecourt.gov.pk</a>                            | Key judgments & precedents |
| FBR / FIA     | <a href="http://fbr.gov.pk">fbr.gov.pk</a> / <a href="http://fia.gov.pk">fia.gov.pk</a> | Tax, cybercrime laws       |

## 6. Challenges & Solutions

**C1: Legal Accuracy — Galat advice dangerous**

HIGH

**Problem:** AI galat legal guidance de toh user ko real harm ho sakta hai.

**Solution:** Har response pe clear disclaimer: 'Yeh general guidance hai, specific cases ke liye lawyer se milein.' Triage system serious cases mein khud lawyer refer karta hai. RAG grounded answers deta hai — hallucination reduce hoti hai.

**C2: Urdu NLP Accuracy**

MEDIUM

**Problem:** LLMs Urdu/Roman Urdu mein English se weak hote hain.

**Solution:** Mistral + extensive Urdu few-shot examples. Roman Urdu normalization layer (unicode conversion). Whisper Urdu speech recognition reliable hai. Extensive testing required.

**C3: Law PDF Structure Inconsistency**

MEDIUM

**Problem:** Government law PDFs poorly formatted, scanned images bhi hote hain.

**Solution:** pdfplumber + OCR pipeline (Tesseract) for scanned docs. Manual chunking for critical laws. LangChain custom document loaders.

**C4: Liability — Galat advice pe blame**

HIGH

**Problem:** Agar user galat action le aur koi nuksan ho toh platform pe blame aa sakta hai.

**Solution:** Terms of Service mein crystal-clear disclaimer. 'AI assistant hai, lawyer nahi' har page pe visible. RED-tier cases mein khud system refuse kare specific advice se aur lawyer refer kare.

**C5: Low-quality voice input**

LOW

**Problem:** Rural users ke basic phones mein mic quality poor, background noise zyada.

**Solution:** Noise reduction preprocessing before Whisper. Text input always available fallback. WhatsApp text channel as alternative.

**C6: Laws update hoti rehti hain**

MEDIUM

**Problem:** Parliament amendments pass karte rehta hai, outdated law give karna dangerous.

**Solution:** Version-controlled law database. Quarterly manual review cycle. Each law chunk mein version/date metadata store hoti hai.

**C7: User Trust — AI pe trust nahi**

LOW

**Problem:** Sensitive legal cases mein log AI pe trust nahi karte.

**Solution:** Anonymous mode (no login, no data). Open source code. NGO partnerships for credibility. Clear source citation (kaunse law section se answer aya).

## 7. Implementation Roadmap

### Phase 1 — Foundation (Month 1-2)

- FastAPI project setup + PostgreSQL schema
- Law PDF ingestion pipeline (PPC, CrPC, Tenancy, Family Laws)
- FAISS vector database setup + LangChain RAG pipeline
- Basic text Q&A; in Urdu (no voice yet)
- Simple web frontend (mobile-first)
- Anonymous session management

### Phase 2 — Voice & Documents (Month 3-4)

- Whisper integration — Urdu speech-to-text
- gTTS — Urdu text-to-speech output
- Roman Urdu normalization layer
- Triage classifier (GREEN/YELLOW/RED)
- Legal document draft generator (FIR, legal notice, consumer complaint)
- Nearest legal aid finder (by city)

### Phase 3 — Advanced Features (Month 5-6)

- Document OCR + fake document detector (Tesseract)
- Punishment & fine estimator
- Emergency rights card generator (printable)
- WhatsApp/SMS integration (Twilio)
- Law update tracking system
- Lawyer referral network (revenue stream)

### Phase 4 — Launch & Scale (Month 7)

- End-to-end testing with real legal scenarios

|   |                                              |
|---|----------------------------------------------|
| ■ | NGO partnership outreach (AGHS, HRCP, Rozan) |
| ■ | HuggingFace Spaces free deployment           |
| ■ | FYP presentation + live demo                 |
| ■ | User feedback collection                     |
| ■ | Grant application (UNDP, Ministry of Law)    |

## 8. Business Model & Sustainability

Core service hamesha free rahegi citizens ke liye — yeh mission hai. Revenue streams platform ko sustainable rakhne ke liye hain.

### Revenue Streams

| Stream            | How                                                                   | Estimated Revenue           |
|-------------------|-----------------------------------------------------------------------|-----------------------------|
| NGO/UNDP Grants   | Apply for legal-aid grants (UNDP, Ford Foundation active in Pakistan) | PKR 100k-5M one-time        |
| Lawyer Referrals  | Complex RED cases mein verified lawyers suggest — per case            | PKR 500-2000/referral       |
| Government Tender | Ministry of Law / HEC partnership for citizen services                | Tender-based contract       |
| Law Firms (B2B)   | Lawyers use our system for quick case research — monthly subscription | PKR 1,500-8,000/mo per firm |
| WhatsApp Business | Business organizations use for employee legal queries — per chatbot   | PKR 10,000-50,000/mo        |

### Competitive Landscape

| Competitor      | What They Do                       | Our Advantage                                  |
|-----------------|------------------------------------|------------------------------------------------|
| Google/Websites | General legal info (English)       | Urdu + voice + Pakistan-specific + interactive |
| Lawyers         | Personalized advice (expensive)    | Free + 24/7 + anonymous + instant              |
| Legal Aid NGOs  | Free advice (physical office only) | Online + voice + scalable to all Pakistan      |
| No competitor   | Voice AI legal guide in Urdu       | We ARE the first — no direct competition       |

### Social Impact Metrics (Target)

|                            |                       |
|----------------------------|-----------------------|
| Year 1 — Users Served      | 50,000+ citizens      |
| Year 1 — Documents Drafted | 10,000+ FIRs/notices  |
| Year 2 — Cities Covered    | All major cities      |
| Year 3 — Language Support  | 5 Pakistani languages |
| Long-term — Lives Impacted | Millions              |

## 9. Why Waqeel AI — Summary



### Real Problem, Real Impact

Pakistan mein 60%+ legally illiterate — yeh sirf ek statistic nahi, yeh rozmarra ke financial aur social harm hai. Waqeel AI directly is harm ko reduce karta hai.



### No Existing Competition

Search results confirm karte hain: koi dedicated free voice-based Urdu legal AI Pakistan mein exist nahi karta. Hum genuinely first hain.



### Technically Impressive for FYP

RAG + Voice (Whisper) + Urdu NLP + OCR + Triage ML + Document Generation — yeh combination examiner panel ke liye genuinely novel hai.



### Fully Deployable

HuggingFace Spaces pe free deploy ho sakta hai — live demo FYP defense mein dikhaya ja sakta hai, sirf slides nahi.



### Sustainable Model

Free for citizens + NGO grants + lawyer referrals = mission-aligned sustainable business. Koi government permission ya approval required nahi.



### Personal Connection

Aap Pakistani hain — aap khud yeh problem jaante hain. Judges aur clients ko yeh authenticity dikhti hai aur trust build karta hai.

**Waqeel AI sirf ek FYP project nahi — yeh Pakistan ke 220 million logo ke liye ek zaruri service hai.**